Deng Zhang

Email：zhangd@whu.edu.cn

EDUCATION BACKGROUND

Ph.D. in School of Water Resources and Hydropower Engineering, Wuhan University 2024-Now

Supervisors: Prof. Jiyun Song

M.Eng. in School of Water Resources and Hydropower Engineering, Wuhan University 2022-2024

Supervisors: Prof. Jie Chen

Outstanding New Graduate Student Scholarship

B.Eng. in Hydrology and Water Resources Engineering, Northwest A&F University 2018-2022

Ranking：2/57 GPA：3.87/4

The National Scholarship once, National Motivational Scholarship twice, Professional Scholarship four times

RESEARCH INTERESTS

- *Thermal Environment Background and Phenomena*: Extreme Heat, Climate Change, Urban Heat Island
- *Integrated Heat Risk and Vulnerability*: Urban Heat Risk and Vulnerability, High Spatiotemporal Resolution
- *Adaptation Strategies and Technical Approaches*: Blue-Green Space Optimization, Ventilation Corridor, Local Cooling Effect

CONFERENCE PRESENTATIONS

1. **Zhang, Deng** and Song, Jiyun*: Machine Learning-Driven Hourly Heat Index Mapping via Multi-Source Data Fusion: Optimizing Blue-Green Infrastructure and Ventilation Corridors for Thermal Resilience in Wuhan, 12th International Conference on Urban Climate, Rotterdam, The Netherlands, 7–11 Jul 2025, ICUC12-165, https://doi.org/10.5194/icuc12-165, 2025.
2. **Zhang, Deng** and Song, Jiyun*: High-Resolution Spatiotemporal Patterns of Urban Humid Heat Risk and Nature-Based Cooling Solutions, The 10th Youth Forum on Geosciences, Hefei, China, 9–13 May 2025.
3. **Zhang, Deng** and Song, Jiyun*: A Study on Urban Heat Risk Mitigation Strategies Based on Blue-Green Spaces and Ventilation Corridor, The 22nd China Water Forum, Xi'an, China, 12–14 Sep 2025.

RESEARCH EXPERIENCE

- **Doctoral Research** 2024.01-Now

Fine-scale urban meteorological information mapping with machine learning, such as temperature, humidity and wind, investigation and mitigation of urban heat island, monitoring and research of urban carbon flux

- **Master's Research** 2022.09-2023.12

Medium- and long-term runoff prediction in the Jinsha River basin: responsible for the evaluation of the accuracy of subseasonal meteorological prediction products, bias correction, hydrological model construction and calibration, and subseasonal runoff prediction by combining meteorology and hydrology

- **Undergraduate Thesis** 2022.02-2022.06

Evaluating the simulation accuracy of the dynamical downscaling data TPReanalysis and the reanalysis data ERA5 and ERA5-land for short-term extreme precipitation on the Tibetan Plateau, the results of the study were awarded as Outstanding Diploma Thesis

TECHNICAL SKILLS

–**High-Performance Computing & Numerical Modeling:** WRF/WRF-UCM (multi-domain nesting, urban canopy schemes),

MPI parallel computing, NetCDF processing; Linux HPC clusters (Slurm/PBS), job scheduling.
–**Machine Learning & Interpretability:** Neural Networks (LSTM), SVM, Random Forest; causal inference; SHAP, partial dependence plots.
– **Scientific Programming:** Python (xarray, matplotlib, scikit-learn, joblib), MATLAB; Origin for data visualization.
– **Geospatial & Microclimate Analysis:** ArcGIS, ENVI-met, satellite/mobile sensor fusion; local climate zone (LCZ) mapping; thermal comfort indices (HI, UTCI).
– **Languages:** English (CET-6), Chinese (Native).